

Numerical models

(2/3): EX 6–10

Outline

EX 6 — Vehicles with a minimum quantity

EX 7 — Custom aircraft with a fixed set-up cost

EX 8 — Seminars

EX 9 — Queens on the chessboard

EX 10 — Tools of a CNC machine

The five parts of every numerical model

- ▶ the statement, with the data of the instance;
- ▶ the **variables**, with their domain and their count;
- ▶ the **model on the instance**, primal and dual;
- ▶ a feasible solution built by hand, which gives the **primal bound**;
- ▶ a dual solution built by hand, which gives the **dual bound**, and the comparison with the optimum of the solver.

EX 6 — Vehicles with a minimum quantity

Introduction: EX 6

- ▶ A production plan with two resources (steel and labour hours)
- ▶ If a vehicle type is selected, a **minimum quantity** of it must be produced
- ▶ The technique: the **semi-integer variable**, obtained with a minimum-quantity constraint and an activation constraint
- ▶ We build a primal and a dual bound by hand and compare them with the optimum

EX 6 — five types of vehicle

A company is considering the production of five vehicle types. It has 1000 tons of steel and 2000 labour hours available.

Type	steel	hours	profit	minimum
compact car	2	30	200	10
midsize car	3	25	250	10
large car	5	40	300	10
midsize minivan	6	45	550	5
large minivan	8	55	700	5

The minimum is the quantity to be produced if the type is selected.
The company wants the plan of maximum profit.

The variables

Five types and two resources: one quantity per type and one indicator per type.

$$x_j = \text{units produced of type } j, \quad \forall j \in \{1, \dots, 5\},$$

$$y_j = \begin{cases} 1 & \text{if type } j \text{ is produced,} \\ 0 & \text{otherwise,} \end{cases} \quad \forall j \in \{1, \dots, 5\}.$$

- The activation coefficient is $M_j = \min\{\lfloor b_i/a_{ij} \rfloor : a_{ij} > 0\}$: on the instance 66, 80, 50, 44 and 36 units, and in every case the limiting resource is the labour time

Description: the semi-integer variable

The minimum-quantity constraint and the activation constraint together make x_j a **semi-integer** variable (the units are whole numbers):

y_j	what the two constraints impose	x_j allowed
0	$x_j \geq 0$ and $x_j \leq 0$	only $x_j = 0$
1	$x_j \geq q_j$ and $x_j \leq M_j$	$q_j \leq x_j \leq M_j$

- ▶ Neither is enough on its own
- ▶ Without the activation a type could be produced with $y_j = 0$
- ▶ Without the minimum quantity the minimum lot would be empty

EX 6 on the instance — primal

EX 6 on the instance — primal

$$\begin{array}{ll}
 \max & 200x_1 + 250x_2 + 300x_3 + 550x_4 + 700x_5 \\
 \text{subject to} & 2x_1 + 3x_2 + 5x_3 + 6x_4 + 8x_5 \leq 1000 \\
 & 30x_1 + 25x_2 + 40x_3 + 45x_4 + 55x_5 \leq 2000 \\
 & x_1 - 10y_1 \geq 0 \\
 & \quad x_2 - 10y_2 \geq 0 \\
 & \quad \quad x_3 - 10y_3 \geq 0 \\
 & \quad \quad \quad x_4 - 5y_4 \geq 0 \\
 & \quad \quad \quad \quad x_5 - 5y_5 \geq 0 \\
 & x_1 - 66y_1 \leq 0 \\
 & \quad x_2 - 80y_2 \leq 0 \\
 & \quad \quad x_3 - 50y_3 \leq 0 \\
 & \quad \quad \quad x_4 - 44y_4 \leq 0 \\
 & \quad \quad \quad \quad x_5 - 36y_5 \leq 0 \\
 & x_1, x_2, x_3, x_4, x_5 \in \mathbb{Z}_{\geq 0} \\
 & \quad y_1, y_2, y_3, y_4, y_5 \in \{0, 1\}
 \end{array}$$

The dual of the LP relaxation

- ▶ A non-negative variable π_i for each resource
- ▶ A non-positive variable λ_j for each minimum-quantity constraint ($\lambda_j = -\ell_j$ with $\ell_j \geq 0$)
- ▶ A non-negative variable μ_j for each activation constraint
- ▶ π_i is the price of one unit of resource i
- ▶ ℓ_j and μ_j are the prices of the two constraints that make x_j semi-integer
- ▶ The objective values all the available resources at those prices

EX 6 on the instance — dual

EX 6 on the instance — dual

$$\begin{array}{llllllllll}
 \min & 1000\pi_1 + 2000\pi_2 & & & & & & & & \\
 \text{subject to} & 2\pi_1 & +30\pi_2 & +\lambda_1 & & & +\mu_1 & & & \geq 200 \\
 & 3\pi_1 & +25\pi_2 & & +\lambda_2 & & & +\mu_2 & & \geq 250 \\
 & 5\pi_1 & +40\pi_2 & & & +\lambda_3 & & & +\mu_3 & \geq 300 \\
 & 6\pi_1 & +45\pi_2 & & & & +\lambda_4 & & & +\mu_4 \geq 550 \\
 & 8\pi_1 & +55\pi_2 & & & & & +\lambda_5 & & +\mu_5 \geq 700 \\
 & & & -10\lambda_1 & & & & & -66\mu_1 & \geq 0 \\
 & & & & -10\lambda_2 & & & & & -80\mu_2 \geq 0 \\
 & & & & & -10\lambda_3 & & & & & -50\mu_3 \geq 0 \\
 & & & & & & -5\lambda_4 & & & & & -44\mu_4 \geq 0 \\
 & & & & & & & -5\lambda_5 & & & & & -36\mu_5 \geq 0 \\
 & \pi_1, & \pi_2 & & & & & & & & & & \geq 0 \\
 & & & \lambda_1, & \lambda_2, & \lambda_3, & \lambda_4, & \lambda_5 & & & & & \leq 0 \\
 & & & & & & & & \mu_1, & \mu_2, & \mu_3, & \mu_4, & \mu_5 \geq 0
 \end{array}$$

A feasible solution: the primal bound

- ▶ Constructive heuristic on the profit per labour hour: $20/3$, 10 , $15/2$, $110/9$ and $140/11$ for the five types
- ▶ One starts from the large minivan ($140/11$) and produces 36 of them, the maximum allowed by the hours
- ▶ One is left with 712 tons of steel but only 20 hours
- ▶ No other type reaches its own minimum quantity: the constructive heuristic stops

$$z(MILP) \geq LB = 25\,200.$$

A dual solution: the dual bound

With $\ell_j = \mu_j = 0$ (the minimum lot is not priced) and a single resource priced at the highest unit price among the types:

$$\bar{\pi}_{\text{steel}} = \max_j \frac{p_j}{a_{1j}} = 100 \Rightarrow 1000 \cdot 100 = 100\,000,$$

$$\bar{\pi}_{\text{hours}} = \max_j \frac{p_j}{a_{2j}} = \frac{140}{11} \Rightarrow 2000 \cdot \frac{140}{11} = \frac{280\,000}{11} \approx 25\,454.5.$$

The better bound comes from the hours:

$$z(\text{MILP}) \leq UB = \frac{280\,000}{11}.$$

The final comparison (1/2)

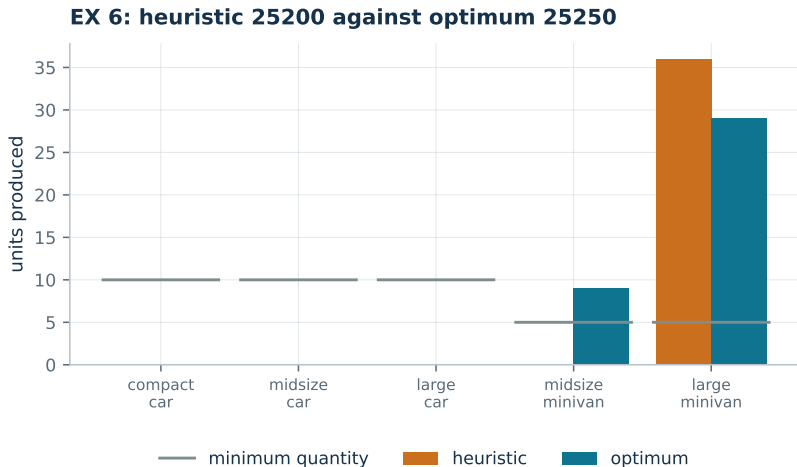
EX 6 vehicles (max)	
LB (heuristic)	25 200
UB (dual by hand)	280000/11
$z(LP)$	280000/11
$z(LP^+)$	229000/9
$z(MILP)$	25 250

- ▶ The optimum produces 9 midsize minivans and 29 large minivans
- ▶ It uses all 2000 hours and 286 tons of steel out of 1000

The final comparison (2/2)

- ▶ The dual bound is **optimal** for the relaxation without the bounds
- ▶ The heuristic is off by a mere 0.2%
- ▶ Without minimum quantities the optimum would be 25 300: the minimum lot costs 50 euros of profit
- ▶ Doubling the minimum quantities one drops to 25 200
- ▶ Doubling the labour hours one reaches 50 750, with the hours still saturated

EX 6: heuristic against optimum



The quantities produced by the heuristic and the optimal ones, with the minimum quantity and the maximum producible of each type.

Summary: EX 6

- ▶ Minimum quantity + activation $\Rightarrow$ **semi-integer** variable:
 $x_j = 0$ or $q_j \leq x_j \leq M_j$
- ▶ The big-M M_j is read from the data: the maximum producible of type j alone
- ▶ Primal bound 25 200 (heuristic on profit per hour), dual bound 280000/11 (a single priced resource)
- ▶ Optimum 25 250: the minimum lot costs 50 euros with respect to the case without minimums

EX 7 — Custom aircraft with a fixed set-up cost

Introduction: EX 7

- ▶ One decides which orders to accept and how many aircraft to produce for each
- ▶ Every accepted order pays a **fixed set-up cost**
- ▶ The technique: a single constraint per client that caps the quantity **and** charges the set-up
- ▶ We will see a case in which the dual bound built by hand settles the problem on its own

EX 7 — which orders to accept

A company builds custom-designed aircraft; before producing for a client a fixed set-up cost has to be paid. Three clients have ordered 3, 2 and 5 aircraft; the production capacity is 1000 hours.

	client 1	client 2	client 3
set-up (millions)	3	2	0
unit profit (millions)	2	3	1
hours per aircraft	200	400	200

Possibly fewer aircraft than requested can be produced. The goal is to maximise the total profit net of the set-up costs.

The variables

- ▶ The data are $n = 3$ clients, with unit profit p_j , set-up cost f_j , hours per aircraft h_j and maximum order M_j
- ▶ The production capacity is $H = 1000$ hours

x_j = aircraft produced for client j (integer),

$$y_j = \begin{cases} 1 & \text{if order } j \text{ is accepted,} \\ 0 & \text{otherwise.} \end{cases}$$

EX 7 on the instance — primal

EX 7 on the instance — primal

$$\begin{array}{llll} \max & 2x_1 & +3x_2 & +x_3 -3y_1 -2y_2 \\ \text{subject to} & 200x_1 +400x_2 +200x_3 & & \leq 1000 \\ & x_1 & & -3y_1 \leq 0 \\ & & x_2 & -2y_2 \leq 0 \\ & & & x_3 -5y_3 \leq 0 \\ & x_1, & x_2, & x_3 \in \mathbb{Z}_{\geq 0} \\ & & & y_1, y_2, y_3 \in \{0, 1\} \end{array}$$

A feasible solution: the primal bound

- ▶ Constructive heuristic on the net profit per hour, computed on the whole order: $3/600$, $4/800$ and $5/1000$, that is $1/200$ for all three
- ▶ With equal ratios one scans in the natural order
- ▶ Client 1 is accepted with 3 aircraft (600 hours, net profit 3)
- ▶ Then client 2 with a single aircraft (400 hours, net profit $3 - 2 = 1$), and the hours run out

$$z(MILP) \geq LB = 4.$$

EX 7 on the instance — dual

EX 7 on the instance — dual

$$\begin{array}{llll} \min & 1000\alpha & & \\ \text{subject to} & 200\alpha + \beta_1 & & \geq 2 \\ & 400\alpha & + \beta_2 & \geq 3 \\ & 200\alpha & & + \beta_3 \geq 1 \\ & & -3\beta_1 & \geq -3 \\ & & & -2\beta_2 \geq -2 \\ & & & & -5\beta_3 \geq 0 \\ & \alpha & & \geq 0 \\ & & \beta_1, & \beta_2, & \beta_3 \geq 0 \end{array}$$

A dual solution: the dual bound

- ▶ The recipe: $\bar{\beta}_j = f_j/M_j$, that is the set-up spread over the aircraft ordered (1, 1 and 0)
- ▶ Hence $\alpha \geq (p_j - \bar{\beta}_j)/h_j$ for every j
- ▶ That is $\alpha \geq 1/200$ in all three cases

With $\bar{\alpha} = 1/200$:

$$z(MILP) \leq UB = 1000 \cdot \frac{1}{200} = 5.$$

The final comparison

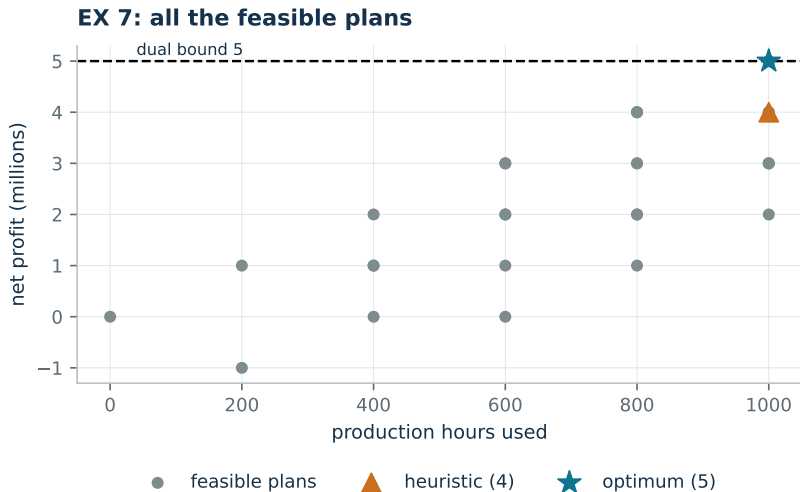
EX 7 aircraft (max)	
LB	4
UB	5
$z(LP)$	5
$z(LP^+)$	5
$z(MILP)$	5

- ▶ The optimum is 5 and is attained by **three** different plans: five aircraft for client 3; two for client 2 and one for the third; three for client 1 and two for the third
- ▶ All three use exactly the 1000 hours

The dual bound settles the problem

- ▶ The dual bound is exact
- ▶ In this case the certificate settles the problem on its own: there is nothing left for the solver to do
- ▶ Without set-up costs the optimum would rise to 9
- ▶ With 1400 hours available, to 7

EX 7: all the feasible plans



Every feasible plan in the hours–profit plane, with the heuristic, the optimum and the dual bound.

Summary: EX 7

- ▶ Fixed cost: the constraint $x_j \leq M_j y_j$ caps the quantity and switches on the set-up $f_j y_j$
- ▶ Primal bound 4 (heuristic on the net profit per hour)
- ▶ Dual bound 5: set-up spread over the aircraft ordered, then a single price for the hour
- ▶ $LB < UB = z(MILP) = 5$: the dual certificate proves optimality without the solver

EX 8 — Seminars

Introduction: EX 8

- ▶ A selection of sessions under constraints of different kinds
- ▶ The technique: an **exact cardinality** (an equality constraint) and **non-adjacency** constraints between consecutive slots
- ▶ In the dual the variable of the equality constraint is **free**
- ▶ We will see that the choice of the dual recipe matters: two recipes give very different bounds

EX 8 — selection with exact cardinality and non-adjacency

A student must attend **exactly two** sessions of a summer school. Three seminars, each offered in four time slots; preference score of each session:

	9–10	10–11	11–12	12–13
Seminar 1	8	6	5	3
Seminar 2	7	9	4	6
Seminar 3	5	7	8	9

- ▶ Every seminar is attended at most once; two sessions cannot occupy the same slot
- ▶ Never **two consecutive hours** of class. The total preference is maximised

The variables

- ▶ The data are $n = 3$ seminars, $m = 4$ slots, $q = 2$ sessions to attend and the score p_{sk}

$$x_{sk} = \begin{cases} 1 & \text{if seminar } s \text{ is attended in slot } k, \\ 0 & \text{otherwise,} \end{cases}$$

- ▶ $n m = 12$ binary variables in total

EX 8 on the instance — primal

EX 8 on the instance — primal

$$\begin{aligned}
 &\max 8x_{11} + 6x_{12} + 5x_{13} + 3x_{14} + 7x_{21} + 9x_{22} + 4x_{23} + 6x_{24} + 5x_{31} + 7x_{32} + 8x_{33} + 9x_{34} \\
 \text{subject to } &x_{11} \qquad \qquad \qquad + x_{21} \qquad \qquad \qquad + x_{31} \qquad \qquad \leq 1 \\
 &\qquad \qquad x_{12} \qquad \qquad \qquad + x_{22} \qquad \qquad \qquad + x_{32} \qquad \qquad \leq 1 \\
 &\qquad \qquad \qquad x_{13} \qquad \qquad \qquad + x_{23} \qquad \qquad \qquad + x_{33} \qquad \leq 1 \\
 &\qquad \qquad \qquad \qquad x_{14} \qquad \qquad \qquad + x_{24} \qquad \qquad \qquad + x_{34} \leq 1 \\
 &x_{11} + x_{12} + x_{13} + x_{14} \qquad \qquad \qquad \qquad \qquad \qquad \leq 1 \\
 &\qquad \qquad \qquad \qquad \qquad x_{21} + x_{22} + x_{23} + x_{24} \qquad \qquad \leq 1 \\
 &\qquad \qquad \qquad \qquad \qquad \qquad \qquad x_{31} + x_{32} + x_{33} + x_{34} \leq 1 \\
 &x_{11} + x_{12} + x_{13} + x_{14} + x_{21} + x_{22} + x_{23} + x_{24} + x_{31} + x_{32} + x_{33} + x_{34} = 2 \\
 &x_{11} + x_{12} \qquad \qquad \qquad + x_{21} + x_{22} \qquad \qquad \qquad + x_{31} + x_{32} \qquad \leq 1 \\
 &\qquad \qquad x_{12} + x_{13} \qquad \qquad \qquad + x_{22} + x_{23} \qquad \qquad \qquad + x_{32} + x_{33} \qquad \leq 1 \\
 &\qquad \qquad \qquad x_{13} + x_{14} \qquad \qquad \qquad + x_{23} + x_{24} \qquad \qquad \qquad + x_{33} + x_{34} \leq 1 \\
 &x_{11}, \ x_{12}, \ x_{13}, \ x_{14}, \ x_{21}, \ x_{22}, \ x_{23}, \ x_{24}, \ x_{31}, \ x_{32}, \ x_{33}, \ x_{34} \in \{0, 1\}
 \end{aligned}$$

A feasible solution: the primal bound

Constructive heuristic: take the feasible session with the highest score, then the best compatible one.

- ▶ **Step 1:** seminar 3 in slot 4 (score 9)
- ▶ **Step 2:** among the sessions that do not use seminar 3, do not occupy slot 4 and are not adjacent to it, the best is seminar 2 in slot 2 (score 9)

In a **maximisation** the value is a **lower** bound:

$$LB = 18.$$

The dual variables

- ▶ $\alpha_k \geq 0$: price of time slot k
- ▶ $\beta_s \geq 0$: price of seminar s
- ▶ γ **free**: price of the exact cardinality constraint
- ▶ $\delta_k \geq 0$: price of the non-adjacency constraint between slots k and $k + 1$

The objective adds up the right-hand sides of those constraints at those prices.

EX 8 on the instance — dual

EX 8 on the instance — dual

$$\begin{aligned}
 & \min \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \beta_1 + \beta_2 + \beta_3 + 2\gamma + \delta_1 + \delta_2 + \delta_3 \\
 & \text{subject to } \alpha_1 \quad \quad \quad + \beta_1 \quad \quad \quad + \gamma + \delta_1 \quad \geq 8 \\
 & \quad \quad \alpha_2 \quad \quad \quad + \beta_1 \quad \quad \quad + \gamma + \delta_1 + \delta_2 \geq 6 \\
 & \quad \quad \quad \alpha_3 \quad + \beta_1 \quad \quad \quad + \gamma \quad + \delta_2 + \delta_3 \geq 5 \\
 & \quad \quad \quad \quad \alpha_4 + \beta_1 \quad \quad \quad + \gamma \quad \quad + \delta_3 \geq 3 \\
 & \alpha_1 \quad \quad \quad + \beta_2 \quad + \gamma + \delta_1 \quad \geq 7 \\
 & \quad \quad \alpha_2 \quad \quad \quad + \beta_2 \quad + \gamma + \delta_1 + \delta_2 \geq 9 \\
 & \quad \quad \quad \alpha_3 \quad \quad \quad + \beta_2 \quad + \gamma \quad + \delta_2 + \delta_3 \geq 4 \\
 & \quad \quad \quad \quad \alpha_4 \quad + \beta_2 \quad + \gamma \quad \quad + \delta_3 \geq 6 \\
 & \alpha_1 \quad \quad \quad \quad \quad + \beta_3 \quad + \gamma + \delta_1 \quad \geq 5 \\
 & \quad \quad \alpha_2 \quad \quad \quad \quad \quad + \beta_3 \quad + \gamma + \delta_1 + \delta_2 \geq 7 \\
 & \quad \quad \quad \alpha_3 \quad \quad \quad \quad \quad + \beta_3 \quad + \gamma \quad + \delta_2 + \delta_3 \geq 8 \\
 & \quad \quad \quad \quad \alpha_4 \quad \quad \quad \quad \quad + \beta_3 \quad + \gamma \quad \quad + \delta_3 \geq 9 \\
 & \alpha_1, \alpha_2, \alpha_3, \alpha_4 \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \geq 0 \\
 & \quad \quad \quad \beta_1, \beta_2, \beta_3 \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \geq 0 \\
 & \quad \quad \quad \quad \quad \quad \gamma \quad \quad \quad \quad \quad \quad \quad \quad \geq 0 \\
 & \quad \quad \quad \quad \quad \quad \delta_1, \delta_2, \delta_3 \geq 0
 \end{aligned}$$

A dual solution: the dual bound

The shortest recipe: everything at zero except $\bar{\gamma} = \max_{s,k} p_{sk} = 9$.

- ▶ Every dual constraint becomes $9 \geq p_{sk}$, true by construction
- ▶ The value is $q\bar{\gamma} = 2 \cdot 9 = 18$
- ▶ It reads: “two sessions are attended, and none is worth more than the best one”

Dual recipe	<i>UB</i>
everything at zero except $\gamma = \max_{s,k} p_{sk}$	18
$\gamma = 0, \alpha_k = \max_s p_{sk}$	34

The recipe matters

- ▶ The second recipe reads “every slot is worth at most its best”
- ▶ The first is optimal for the relaxation
- ▶ The second, apparently more natural, is almost twice as large: 34 against 18

The recipe matters.

The final comparison

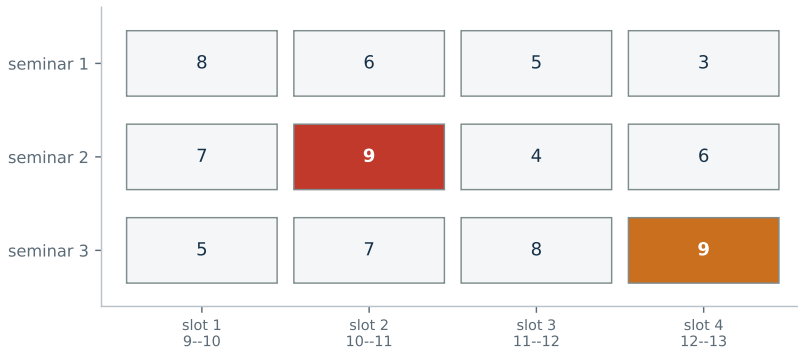
The optimum is $z(MILP) = 18$: seminar 2 in slot 2 and seminar 3 in slot 4, with a break in between, as required.

LB (constructive heuristic)	18
$z(MILP)$	18
$z(LP)$	18
UB (dual by hand)	18
heuristic gap	0.0%

The two hand-built bounds **close** the problem: $z(MILP) = 18$ is proved without the solver.

EX 8: the two chosen sessions

EX 8: the two chosen sessions ($z = 18$)



The grid of preferences; highlighted, the two chosen sessions.

Summary: EX 8

- ▶ Exact cardinality $\sum x_{sk} = q$: in the dual its variable γ is **free**
- ▶ Non-adjacency: one constraint for each pair of consecutive slots
- ▶ Primal bound 18 (greedy heuristic), dual bound 18 ($\bar{\gamma} = 9$, everything else at zero)
- ▶ Different dual recipes give very different bounds (18 against 34): the recipe matters

EX 9 — Queens on the chessboard

Introduction: EX 9

- ▶ The classical queens problem, on a 4×4 chessboard
- ▶ The technique: a **set packing** on rows, columns and diagonals
- ▶ The indexing trick for the diagonals: $i - j$ and $i + j$ constant
- ▶ The dual is a **set covering**, and a very simple certificate proves optimality

EX 9 — the largest number of queens

- ▶ On a 4×4 chessboard one wants to place the largest number of queens so that no queen can capture another in a single move
- ▶ In chess a queen moves any number of squares along the same row, the same column or either diagonal

The variables

One binary variable per square of the board:

$$x_{ij} = \begin{cases} 1 & \text{if there is a queen on square } (i,j), \\ 0 & \text{otherwise,} \end{cases}$$

$$\forall i \in \{1, 2, 3, 4\}, \forall j \in \{1, 2, 3, 4\}.$$

- Sixteen columns, one per square of the board

EX 9 on the instance — primal

EX 9 on the instance — primal

$$\begin{array}{rcl}
\max & x_{11} + x_{12} + x_{13} + x_{14} + x_{21} + x_{22} + x_{23} + x_{24} + x_{31} + x_{32} + x_{33} + x_{34} + x_{41} + x_{42} + x_{43} + x_{44} & \\
\text{subject to} & x_{11} + x_{12} + x_{13} + x_{14} & \leq 1 \\
& x_{21} + x_{22} + x_{23} + x_{24} & \leq 1 \\
& x_{31} + x_{32} + x_{33} + x_{34} & \leq 1 \\
& x_{41} + x_{42} + x_{43} + x_{44} & \leq 1 \\
& x_{11} & + x_{21} & + x_{31} & + x_{41} & \leq 1 \\
& x_{12} & & + x_{22} & & + x_{32} & & + x_{42} & \leq 1 \\
& & x_{13} & & + x_{23} & & + x_{33} & & + x_{43} & \leq 1 \\
& & & x_{14} & & + x_{24} & & + x_{34} & & + x_{44} \leq 1 \\
& & & & x_{13} & & + x_{24} & & & \leq 1 \\
& & & & & x_{12} & & + x_{23} & & + x_{34} & \leq 1 \\
& & & & & & x_{11} & & + x_{22} & & + x_{33} & & + x_{44} \leq 1 \\
& & & & & & & x_{21} & & & + x_{32} & & + x_{43} & \leq 1 \\
& & & & & & & & x_{31} & & & + x_{42} & \leq 1 \\
& & & & & & & & & x_{12} & + x_{21} & \leq 1 \\
& & & & & & & & & & x_{13} & + x_{22} & + x_{31} & \leq 1 \\
& & & & & & & & & & & x_{14} & + x_{23} & + x_{32} & + x_{41} & \leq 1 \\
& & & & & & & & & & & & x_{24} & + x_{33} & + x_{42} & \leq 1 \\
& & & & & & & & & & & & & x_{34} & + x_{43} & \leq 1 \\
& & & & & & & & & & & & & & x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34}, x_{41}, x_{42}, x_{43}, x_{44} \in \{0, 1\}
\end{array}$$

Description

- ▶ A **set packing** on four families of lines: the 4 rows, the 4 columns and the diagonals in both directions
- ▶ 18 constraints in total
- ▶ The indexing trick is all here:
 - ▶ the “descending” diagonals are the sets with $i - j$ constant
 - ▶ the “ascending” ones the sets with $i + j$ constant

Observation 1: Diagonals with a single square do not become constraints

Each of the four corners sits on a diagonal that touches nothing else. Writing its constraint would give $x_{ij} \leq 1$ on an already binary variable: a row that forbids nothing. What remains is $5 + 5$ diagonals with at least two squares, and the model loses ten rows without losing a single solution.

A feasible solution: the primal bound

Constructive heuristic row by row:

- ▶ in row i one picks the first free column that is not attacked by the queens already placed; if there is none, the row stays empty
- ▶ it never backtracks

On the instance:

- ▶ queens on $(1, 1)$ and $(2, 3)$
- ▶ row 3 has no free square left and is skipped
- ▶ in row 4 it takes $(4, 2)$

$$z(MILP) \geq LB = 3.$$

EX 9 on the instance — dual

EX 9 on the instance — dual

$$\begin{aligned}
 & \min \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \beta_1 + \beta_2 + \beta_3 + \beta_4 + \gamma_{-1} + \gamma_0 + \gamma_1 + \gamma_2 + \gamma_3 + \delta_2 + \delta_3 + \delta_4 + \delta_5 + \delta_6 \\
 & \text{subject to} \\
 & \alpha_1 \quad \quad \quad + \beta_1 \quad \quad \quad + \gamma_1 \quad \quad \quad \geq 1 \\
 & \alpha_1 \quad \quad \quad + \beta_2 \quad \quad \quad + \gamma_0 \quad \quad \quad + \delta_2 \quad \quad \geq 1 \\
 & \alpha_1 \quad \quad \quad + \beta_3 \quad \quad \quad + \gamma_{-1} \quad \quad \quad + \delta_3 \quad \quad \geq 1 \\
 & \alpha_1 \quad \quad \quad + \beta_4 \quad \quad \quad + \delta_4 \quad \quad \geq 1 \\
 & \alpha_2 \quad \quad + \beta_1 \quad \quad \quad + \gamma_2 \quad \quad + \delta_2 \quad \geq 1 \\
 & \alpha_2 \quad \quad + \beta_2 \quad \quad \quad + \gamma_1 \quad \quad + \delta_3 \quad \geq 1 \\
 & \alpha_2 \quad \quad + \beta_3 \quad \quad \quad + \gamma_0 \quad \quad + \delta_4 \quad \geq 1 \\
 & \alpha_2 \quad \quad + \beta_4 + \gamma_{-1} \quad \quad \quad + \delta_5 \quad \geq 1 \\
 & \alpha_3 \quad + \beta_1 \quad \quad \quad + \gamma_3 \quad \quad + \delta_3 \quad \geq 1 \\
 & \alpha_3 \quad + \beta_2 \quad \quad \quad + \gamma_2 \quad \quad + \delta_4 \quad \geq 1 \\
 & \alpha_3 \quad + \beta_3 \quad \quad \quad + \gamma_1 \quad \quad + \delta_5 \quad \geq 1 \\
 & \alpha_3 \quad + \beta_4 \quad \quad + \gamma_0 \quad \quad \quad + \delta_6 \geq 1 \\
 & \alpha_4 + \beta_1 \quad \quad \quad + \delta_4 \quad \geq 1 \\
 & \alpha_4 + \beta_2 \quad \quad \quad + \gamma_3 \quad \quad + \delta_5 \geq 1 \\
 & \alpha_4 + \beta_3 \quad \quad \quad + \gamma_2 \quad \quad + \delta_6 \geq 1 \\
 & \alpha_4 + \beta_4 \quad \quad + \gamma_1 \quad \quad \geq 1 \\
 & \alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0 \\
 & \beta_1, \beta_2, \beta_3, \beta_4 \geq 0 \\
 & \gamma_{-1}, \gamma_0, \gamma_1, \gamma_2, \gamma_3 \geq 0 \\
 & \delta_2, \delta_3, \delta_4, \delta_5, \delta_6 \geq 0
 \end{aligned}$$

A dual solution: the dual bound

For example one can use a single line:

- ▶ $\bar{\alpha}_i = 1$ for every row, everything else zero
- ▶ Every square (i, j) has $\alpha_i = 1 \geq 1$, so the solution is feasible and is worth 4
- ▶ It is the dual translation of the sentence “at most one queen sits in each row”

$$z(MILP) \leq UB = 4.$$

The final comparison

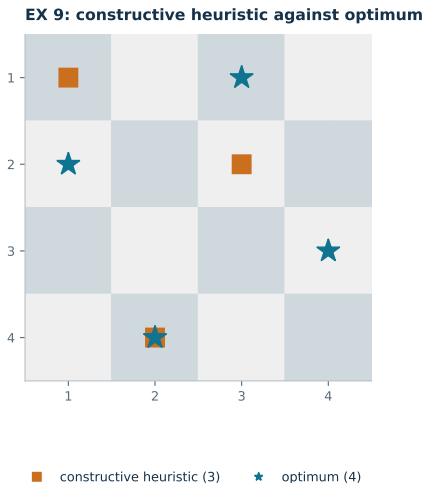
EX 9 queens (max)	
LB	3
UB	4
$z(LP)$	4
$z(LP^+)$	4
$z(MILP)$	4

- ▶ The dual bound is attained: the solution found is optimal, and one knows it without trusting the solver
- ▶ The constructive heuristic, on the other hand, stops at 3: it is the heuristic, not the bound, that leaves the 25% gap

Two variants

- ▶ On $n \times n$ boards with $n = 5$ and $n = 6$ the optimum is always n , and the certificate $\bar{\alpha}_i = 1$ proves it
- ▶ For $n = 2$ and $n = 3$ the optimum is 1 and 2: the bound n stays valid but is not attainable
- ▶ The 4×4 board is the smallest on which n queens actually fit, and therefore the smallest on which the certificate settles it
- ▶ Removing the diagonal constraints one gets **rooks**: the model becomes an assignment, the matrix is totally unimodular and the LP relaxation already gives an integer value

EX 9: heuristic against optimum



The three queens placed by the heuristic and the four of the optimal solution.

Summary: EX 9

- ▶ Queens = **set packing** on rows, columns and diagonals ($i - j$ and $i + j$ constant)
- ▶ Only the diagonals with at least two squares become constraints: 18 in total
- ▶ Primal bound 3 (row-by-row heuristic), dual bound 4 ($\bar{\alpha}_i = 1$)
- ▶ The dual is a set covering; the certificate settles the problem for $n \geq 4$

EX 10 — Tools of a CNC machine

Introduction: EX 10

- ▶ One chooses the operations to perform and the tools to load into a limited magazine
- ▶ The technique: the **disaggregated activation**, one constraint for every operation–tool pair
- ▶ We will see that a dual solution is to be **checked**, not proposed
- ▶ And a case in which the relaxation is very weak

EX 10 — selection with a tool set

A CNC machine can perform five operations; each one requires a subset of six tools and, if performed, yields a profit:

Operation	Required tools	Profit (€)
1	1, 3, 4, 5	2000
2	1, 2, 6	1500
3	1, 2, 3, 5	1800
4	2, 4, 5	1700
5	5, 6	800

The magazine holds at most $k = 4$ tools at a time. The configuration of maximum profit is wanted.

The variables

- ▶ The data are $n = 5$ operations of profit p_i , $m = 6$ tools and a magazine holding $k = 4$ tools
- ▶ T_i is the set of tools required by operation i

$$x_i = \begin{cases} 1 & \text{if operation } i \text{ is performed,} \\ 0 & \text{otherwise,} \end{cases} \quad (n \text{ binaries})$$

$$y_j = \begin{cases} 1 & \text{if tool } j \text{ is loaded,} \\ 0 & \text{otherwise,} \end{cases} \quad (m \text{ binaries})$$

EX 10 on the instance — primal

EX 10 on the instance — primal

$$\begin{array}{ll}
 \max & 2000x_1 + 1500x_2 + 1800x_3 + 1700x_4 + 800x_5 \\
 \text{subject to} & y_1 + y_2 + y_3 + y_4 + y_5 + y_6 \leq 4 \\
 & x_1 \quad \quad \quad -y_1 \leq 0 \\
 & x_1 \quad \quad \quad -y_3 \leq 0 \\
 & x_1 \quad \quad \quad -y_4 \leq 0 \\
 & x_1 \quad \quad \quad -y_5 \leq 0 \\
 & \quad x_2 \quad \quad -y_1 \leq 0 \\
 & \quad x_2 \quad \quad -y_2 \leq 0 \\
 & \quad x_2 \quad \quad -y_6 \leq 0 \\
 & \quad \quad x_3 \quad -y_1 \leq 0 \\
 & \quad \quad x_3 \quad -y_2 \leq 0 \\
 & \quad \quad x_3 \quad -y_3 \leq 0 \\
 & \quad \quad x_3 \quad -y_5 \leq 0 \\
 & \quad \quad \quad x_4 \quad -y_2 \leq 0 \\
 & \quad \quad \quad x_4 \quad -y_4 \leq 0 \\
 & \quad \quad \quad x_4 \quad -y_5 \leq 0 \\
 & \quad \quad \quad \quad x_5 \quad -y_5 \leq 0 \\
 & \quad \quad \quad \quad x_5 \quad -y_6 \leq 0 \\
 & x_1, \quad x_2, \quad x_3, \quad x_4, \quad x_5 \in \{0, 1\} \\
 & y_1, y_2, y_3, y_4, y_5, y_6 \in \{0, 1\}
 \end{array}$$

Description

- ▶ One magazine constraint and $\sum_i |T_i| = 4 + 3 + 4 + 3 + 2 = 16$ link constraints
- ▶ It is the **disaggregated** activation read the other way round: here the operation is the object and the tools are the resources
- ▶ The operation needs **all** of its tools: a conjunction in the consequent, which costs one constraint per tool

A feasible solution: the primal bound

Constructive heuristic by decreasing profit: the operations are scanned from the most profitable and the tool set of each is loaded, if the magazine allows it.

- ▶ **Operation 1** (2000, tools 1, 3, 4, 5): 4 new tools are needed and the magazine reaches exactly 4: it runs
- ▶ **Operations 3, 4, 2, 5**: each would need at least one more tool, and the magazine is full: they are discarded

In a **maximisation** the value is a **lower** bound:

$$LB = 2000.$$

EX 10 on the instance — dual

EX 10 on the instance — dual

min 4α

subject to $\beta_{11} + \beta_{13} + \beta_{14} + \beta_{15} \geq 2000$

$\beta_{21} + \beta_{22} + \beta_{26} \geq 1500$

$\beta_{31} + \beta_{32} + \beta_{33} + \beta_{35} \geq 1800$

$\beta_{42} + \beta_{44} + \beta_{45} \geq 1700$

$\beta_{55} + \beta_{56} \geq 800$

$-\alpha + \beta_{11} \quad \quad \quad + \beta_{21} \quad \quad \quad + \beta_{31} \quad \quad \quad \leq 0$

$-\alpha \quad \quad \quad \quad \quad + \beta_{22} \quad \quad \quad + \beta_{32} \quad \quad \quad + \beta_{42} \quad \quad \quad \leq 0$

$-\alpha \quad \quad + \beta_{13} \quad \quad \quad \quad \quad \quad \quad + \beta_{33} \quad \quad \quad \leq 0$

$-\alpha \quad \quad \quad + \beta_{14} \quad \quad \quad \quad \quad \quad \quad + \beta_{44} \quad \quad \quad \leq 0$

$-\alpha \quad \quad \quad \quad \quad + \beta_{15} \quad \quad \quad \quad \quad + \beta_{35} \quad \quad \quad + \beta_{45} + \beta_{55} \leq 0$

$-\alpha \quad \quad \quad \quad \quad \quad \quad + \beta_{26} \quad \quad \quad \quad \quad + \beta_{56} \leq 0$

$\alpha \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \geq 0$

$\beta_{11}, \beta_{13}, \beta_{14}, \beta_{15}, \beta_{21}, \beta_{22}, \beta_{26}, \beta_{31}, \beta_{32}, \beta_{33}, \beta_{35}, \beta_{42}, \beta_{44}, \beta_{45}, \beta_{55}, \beta_{56} \geq 0$

Description

- ▶ β_{ij} is the share of the profit of operation i attributed to tool j
- ▶ The first group asks the shares to cover the whole profit
- ▶ The second asks that no tool receives more than α
- ▶ The objective pays α for each of the k slots in the magazine

A dual solution: the recipe

- ▶ The profit of each operation is spread in equal parts over the tools it needs:

$$\bar{\beta}_{ij} = \frac{p_i}{|T_i|}$$

the first group is satisfied with equality by construction

- ▶ Then one takes the load of the most demanded tool:

$$\bar{\alpha} = \max_j \sum_{i: j \in T_i} \bar{\beta}_{ij}$$

A dual solution: the dual bound

$\bar{\beta}_{ij} = p_i / T_i $	load of the tool					
	1	2	3	4	5	6
op. 1: 500	500		500	500	500	
op. 2: 500	500	500				500
op. 3: 450	450	450	450		450	
op. 4: 1700/3		1700/3		1700/3	1700/3	
op. 5: 400					400	400
total	1450	4550/3	950	3200/3	5750/3	900

The critical tool is number 5, required by four operations out of five: $\bar{\alpha} = 5750/3$ and

$$UB = k \bar{\alpha} = 4 \cdot \frac{5750}{3} = \frac{23000}{3} \approx 7666.7.$$

Warning (1/2)

Observation 2: A dual solution is to be checked, not proposed

A choice that comes naturally is $\bar{\alpha} = 2000$ with all $\bar{\beta}_{ij} = 900$: the constraints on the operations are satisfied ($2 \cdot 900 = 1800 \geq 800$ is the tightest case) and the value would be $4 \cdot 2000 = 8000$.

***But that solution is not feasible.** The second group asks $\sum_{i:j \in T_i} \beta_{ij} \leq \alpha$, and tool 5 serves **four** operations: it would receive $4 \cdot 900 = 3600 > 2000$. The violation is 1600.*

Warning (2/2)

Observation 3: A dual solution is to be checked, not proposed

The number 8000 is not a bound: it does not come from any feasible dual solution.

*This is why the course protocol does not ask to **propose** a dual solution but to **check** it: the script verifies every constraint, and an assert fails if the violation exceeds the tolerance.*

*With the correct recipe the bound is $23000/3 \approx 7666.7$, that is, **better** than the one hoped for by getting it wrong.*

The final comparison

$z(MILP) = 2500$: tools 2, 4, 5, 6 are loaded and operations 4 (1700) and 5 (800) are performed.

LB (constructive heuristic)	2000
$z(MILP)$	2500
$z(LP)$	5200
$z(LP^+)$	5200
UB (dual by hand)	23000/3
heuristic gap	20.0%

No four-tool configuration yields more: with $\{1, 3, 4, 5\}$ only operation 1 runs (2000), with $\{1, 2, 3, 5\}$ only operation 3 (1800).

Why the relaxation is so weak here (1/2)

- ▶ $z(LP) = 5200$ against $z(MILP) = 2500$: more than double
- ▶ In the continuous problem one can load “a bit” of every tool: for instance $y_j = 2/3$ for six tools, which respects $\sum_j y_j \leq 4$
- ▶ and run a fraction of **every** operation
- ▶ The integrality of the magazine is everything: this is the case where hand-built bounds, however well constructed, are not enough and the work has to be done by the solver

Why the relaxation is so weak here (2/2)

Two ways to tighten the model:

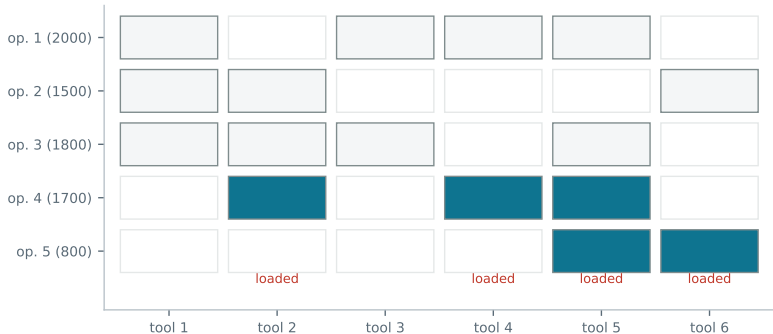
- ▶ add the valid inequalities (the aggregated form of the links)

$$\sum_{j \in T_i} y_j \geq |T_i| x_i \quad \forall i$$

- ▶ observe that any operation with $|T_i| > k$ can be fixed to zero

EX 10: the optimal configuration

EX 10: the operations run (teal) and the 4 tools loaded (z = 2500)



The operation-tool matrix. Highlighted, the operations performed; the annotated columns are the four loaded tools.

Summary: EX 10

- ▶ Disaggregated activation: $x_i \leq y_j$ for every $j \in T_i$ (16 links) plus the magazine constraint
- ▶ Primal bound 2000, optimum 2500, dual bound 23000/3 (profit spread over the tools)
- ▶ A dual solution must be **checked**: 8000 looked like a bound but is not one
- ▶ Weak relaxation (5200): the integrality of the magazine is what matters

Summary (1/2)

Modelling techniques:

- ▶ EX 6: semi-integer variable (minimum quantity + activation)
- ▶ EX 7: fixed cost with a single link constraint per client
- ▶ EX 8: exact cardinality and non-adjacency
- ▶ EX 9: set packing on rows, columns and diagonals
- ▶ EX 10: disaggregated operation–tool activation

Summary (2/2)

Bounds built by hand:

- ▶ In a maximisation the heuristic gives a **lower** bound, a feasible dual solution an **upper** bound
- ▶ In EX 7, EX 8 and EX 9 the dual bound is attained and proves the optimum without the solver
- ▶ The dual recipe matters (EX 8), and every dual solution must be checked (EX 10)
- ▶ When the relaxation is weak (EX 10) the hand-built bounds are not enough